SVKM's NMIMS · Mukesh Patel School of Technology Management & Engineering, Mumbai

Modern Day Robotics & Its Industrial Applications (MDRIIA - 702CO0E012) · B.Tech Computer Science & Business Systems (CSBS), in partnership with TCS · AY 2026–27 Odd Semester

GROUP 06 TECHNICAL DOSSIER

Capstone ICA (20 marks): CDIO Robotics Engineering & MuJoCo Physics 10 · CSBS Techno-Economic Business Systems Analysis 5 · Studio Showcase & Viva Voce 5

EDITABLE REVIEW COPY — unsigned.

Group · project	Group 06 · Autonomous Crawler Cleaning Robot for Inclined Rooftop Solar Panels
Monorepo folder	https://github.com/sunny-nanade/NMIMS-MPSTME-MDRIIA-2026/tree/main/Group_06_Arush_Ronit_Harshvardhan
Submit (3 places)	(1) PDF on MS Teams, turned in by EVERY member · (2) this filled Word file AND its PDF committed to Group_06_Arush_Ronit_Harshvardhan/docs/MDRIIA_Group_06_Technical_Dossier.docx / .pdf · (3) LinkedIn video URL in Section 5. Due Monday, 26-Oct-2026, 23:59 IST. Studio Showcase, Expo & Viva Voce: Saturday, 31-Oct-2026.
No slides	There is no PPT for MDRIIA. At the Expo and in your video you present the live MuJoCo simulation, your business sensitivity graphs, and a webcam picture-in-picture of every member.
How to fill this file	Use Microsoft Word (desktop). Choose answers from the 'Select ▾' drop-downs and type into the 'type here' fields. Do not delete or retype them and do not edit this file in Google Docs: the fields are how your answers are read.

Student name	Roll No.	SAP ID	Capstone role(s)	Specialisation (finalised titles sheet / monorepo)
Arush Ashish Patil	E048	70362400013	Lead Robotics Systems Architect	Lead Tracked Crawler Chassis & MuJoCo Adhesion Modeler
Ronit Rajput	E052	70362400001	Telemetry QA Lead	Waterless Rotary Brush Actuation & Cleaning Efficiency Engineer
Harshvardhan Sahi	E058	70362400031	CSBS Business & Techno-economic Systems Analyst	CSBS Photovoltaic Degradation & CapEx/OpEx Payback Analyst

Roll numbers and SAP IDs are from the official student list. Tell Dr. Nanade by Tuesday, 20-Oct-2026 if any name, ID or role is wrong. Groups with fewer than four members combine roles as shown.

SECTION 1 · Problem Bottleneck, Industry Standards & Aalborg Research Title *Component 1 · CDIO Robotics (10)*

Authorised Aalborg title	How can an autonomous crawler cleaning robot simulated in MuJoCo recover soiling-induced energy losses (15-18% monthly) on inclined commercial rooftop solar arrays while reducing cleaning cycle operational expenditure compared to manual labor?
Original draft topic	Solar Panel Cleaning
Why the title changed	The student submission demonstrated strong local domain awareness (commercial rooftop solar installations in Mumbai). However, it lacked specific empirical parameters for monthly dust soiling attenuation and did not identify the robotic locomotion mechanism (tracked crawler, suction/vacuum adhesion, rotating microfiber brush) or the specific physics simulator.

1.1 Baseline friction: claim verification table

Each statement was carried over from your titles sheet or your monorepo scripts. Before you use it anywhere (dossier, video, paper), find the primary source yourself and record it, or correct or delete the claim. Statistics: cite the original study, not a blog or an AI answer.

Claim (as carried over)	Faculty pre-audit note	Primary source you verified (author, year, venue, DOI/URL)	Source type	Keep / Correct / Delete
A 100 kWp commercial rooftop PV array in Mumbai generates about 140,000 kWh annually.	*≈1,400 kWh/kWp/year is plausible. Confirm with a solar-resource tool or national data for Mumbai and cite it.*	No verified site-specific yield source obtained; delete numerical Mumbai claim. No electrical PV model is implemented.	No source found	Delete
A 16% monthly soiling loss forfeits 22,400 kWh/year.	*Inconsistent: 22,400 kWh is 16% of the annual yield, i.e. an annual-average loss, not a monthly one. Soiling loss depends on the cleaning interval; find a primary Indian soiling study.*	No verified primary monthly-loss source obtained. Monthly loss cannot be multiplied by annual generation as an annual-average loss; delete.	No source found	Delete
Daily waterless cleaning increases net energy yield by 15.2% with CapEx payback within 8.6 months.	*Must come from your own script output with sourced inputs, or be deleted.*	No measured PV electrical recovery or sourced commercial payback supports these numbers. Delete; replace with explicitly assumed dimensionless scenario in Section 3.	No source found	Delete

1.2 Governing industry standards

Decide for each standard whether it governs your system, and name the clause or requirement you design against (cite number, part and year). 'No' is a correct answer when you can say why. The hint is the faculty view; you may disagree with a reason.

Standard	Scope	Faculty hint for your project	Applies?	Clause / requirement you use, or why not
ISO 10218-1/-2	Safety requirements for industrial robots (robot, and robot integration)	*No*	No	Scope: industrial robot/manipulator and integration. This fixed-panel cleaning mobile device is not an industrial manipulator; no conformity claim.
ISO 3691-4	Driverless industrial trucks / AMRs and their systems: safety requirements	*No*	No	Scope: driverless industrial trucks. The selected fixed PV-bay cleaner is not an industrial truck; no conformity claim.
ANSI/RIA R15.06	US national adoption of ISO 10218 (industrial robot safety)	*No*	No	Industrial robot-system standard associated with ISO 10218; selected system is not an industrial manipulator. No compliance claim.
IEC 60601-1	Medical electrical equipment: basic safety and essential performance	*No*	No	Medical electrical equipment; the PV cleaner has no medical use. No applicable medical requirement.
Domain-specific	IEC 61215 / IEC 61730 (PV module qualification and safety) for allowable loads on the glass	*Recommended: check it*	Partly	IEC 61215 (PV module qualification) and IEC 61730 (module safety) are relevant to future module/cleaner compatibility. Licensed clauses and allowable local brush/wheel loads were not verified; no clause number, certified glass load or compliance is asserted.

1.3 Documented baseline friction (who struggles, with what, how it is measured today)

Baseline metric and its value today (with unit)	Measured simulation baseline: waypoint + proportional wheel control; compare at equal 0–30 s exposure. Commercial reference: 120 person-hours/month is an assumption, not a site measurement.
Source of the baseline value (author, year, DOI/URL)	Actual controller logs and frozen protocol; commercial assumptions in analytics/final_business_inputs.json. No Mumbai field survey or verified annual-yield figure.
Who carries the cost today (role / stakeholder)	Rooftop PV owner/operator and cleaning contractor; access, inspection, transfer and maintenance remain human responsibilities in the business scenario.

SECTION 2 · MuJoCo Rigid-Body Physics Modelling & Python Closed-Loop Control *Component 1 · CDIO Robotics (10)*

MuJoCo mechanism (titles sheet)	MuJoCo MJCF Rigid-Body Model with Autonomous Crawler Cleaning Robot for Inclined Rooftop Solar Panels Physics & Closed-Loop Control
MJCF model	Group_06_Arush_Ronit_Harshvardhan/models/collection_cleaning_crawler.xml
Controller	Group_06_Arush_Ronit_Harshvardhan/src/crawler_cleaning_controller.py
Contact & solver settings in the model today	Final model: 0.002 s, five actuators (four driven rollers approximating tracks + one rotary brush). Brush capsule contact priority 2; separate stiff solid contact; zero-friction physical containment guards. See final parameter controls below.
Core state machine (FSM)	Monitor actual RGB → persistent dirt trigger → sweep + requested bottom-edge task → image verification/revisit → return → settle → dock service. No endless loop on a clean scene.
Equations of motion & control	Rigid-body dynamics: $M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = \tau + J^T f$. Ideal uphill no-slip bound: $\mu mg\cos\theta \geq mg\sin\theta$, with Fadh=0. Wheel PI drives four roller motors; heading error creates differential wheel-speed references. Coverage is qualifying brush footprint / 24 m². Dust disappearance uses a contact-gated reduced-order removal model, not granular dust/airflow dynamics.

Audit correction: inherited 16–16.8% soiling and 15.2–16.2% restoration constants are excluded from the final analysis; there is no PV electrical model.

2.1 Model and controller settings you actually used (from your final code)

MJCF timestep (s) in the final model	0.002 s; 500 Hz integration and logging
Main contact friction (sliding / torsional / rolling)	Rollers 1.2 / 0.005 / 0.0001; glass and brush 0.4 / 0.005 / 0.0001; brush contact priority 2. Guard friction is zero. These are model assumptions, not measured tyre/glass properties.
solref (timeconst dampratio) and why you chose it	Brush/glass 0.02 s, 1 (soft bristle approximation); guide 0.01 s, 1; model default 0.005 s, 1. Rounded capsule contact was retained after replay rejected rigid-cylinder kicks. No coating-safety inference.
Controller law used	PI / PD
Final gains (e.g. Kp, Ki, Kd or λ) with units	Wheel PI: Kp=0.8 N·m/(rad/s), Ki=0.5 N·m/rad, integral clamp ±3 rad, motor ±3 N·m. Heading 3 s^{-1}, yaw command ±1.2 rad/s. Shared incline/return regulator Kp=0.6, Ki=0.2; torque ±1 N·m, slew 4 N·m/s. Brush velocity target 15 rad/s, kv=0.1, torque ±1 N·m.
FSM states implemented in code (list)	HOME_MONITOR → SETTLE/ALIGN → SWEEP; image-requested EDGE_CLEAN_BOTTOM; verification/revisit; RETURN_HOME → DOCKING → DOCK_SERVICE → DOCKED. Safety abort/timeout remain failed outcomes.
Model loads and runs: python -c "import mujoco; mujoco.MjModel.from_xml_path(...)"	Yes

2.2 Authorship of each script

Several scripts were provided as starter scaffolds. Mark honestly what you wrote, changed, or used as given. Viva questions target the code you claim.

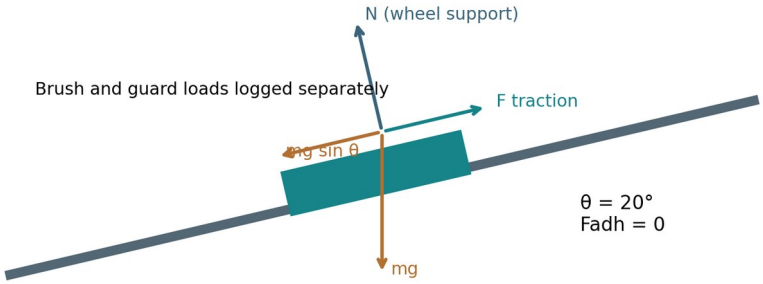
File	Written / Modified / Used as given	Main author (Roll)	What you changed or added (1 line)
models/ collection_cleaning_crawler.xml	Modified by us	E048	Off-glass dock, fixed capsule brush, guide and tray; 8.5 kg and five actuators. Arush led AI-assisted file implementation; Ronit supported 3D modelling.
src/crawler_cleaning_controller.py	Modified by us	E048	Camera-triggered departure, PI tracking, bounded turns/return and 500 Hz logging. Arush led AI-assisted code implementation; team members supported development.
analytics/final_business_model.py	Modified by us	E048	Measured-throughput business scenario, transfer labour and sensitivity. Arush led AI-assisted file implementation; Harshvardhan led business/economic inputs with Ronit supporting business modelling.
src/vision_mission.py + panel_vision.py	Modified by us	E048	Rendered RGB observations, persistence, targeted revisits and independent residual scoring. Arush led AI-assisted implementation; navigation uses simulator pose.
src/run_final_validation.py	Modified by us	E048	Frozen 180 attempts over three inclinations with independent seeds, raw torque logs and source snapshots. Arush led AI-assisted implementation; team members supported the project.

2.3 Key code snippet (≤ 25 lines) implementing your FSM or control law:

src/crawler_cleaning_controller.py, lines 180–185

```
        for i,j in enumerate(joints):
            e=speeds[i]-d.qvel[j]
            integral[i]=np.clip(integral[i]+e*dt,-3,3)
            d.ctrl[i]=np.clip(.8*e+(.5*integral[i] if variant=="proposed" else 0),-3,3)
    d.ctrl[4]=15 if state in ('SWEEP','EDGE_CLEAN_BOTTOM') else 0
    mujoco.mj_forward(m,d)
```

2.4 Free-body diagram and one equation worked with your own numbers and units:



Panel-tangent free-body idealisation; guard and brush contact are distinct from wheel normal load.

m=8.5 kg, g=9.81 m/s², θ=20°: mg=83.385 N; mg sinθ=28.519 N; mg cosθ=78.355 N; ideal μmin=tan20°=0.364. No suction adhesion is simulated. These equilibrium values do not prove traction or glass safety. Actual brush normal load is brush_normal_N, guard load is guard_normal_N and panel velocities are vx/vy_panel_mps in raw CSV. Torque effort is $\Sigma\tau^2\Delta t$; positive mechanical work is $\int \max(\tau\dot{q},0)dt$. Electrical energy additionally requires declared efficiency and auxiliary-power assumptions.

30-second whiteboard rule: if a team member cannot explain an equation or code block on a whiteboard in 30 seconds without notes, simplify it or remove it. It will be asked in the viva.

SECTION 3 · CSBS Techno-Economic Business Systems Modelling *Component 2 · CSBS Techno-Economic (5)*

Business script	Group_06_Arush_Ronit_Harshvardhan/analytics/final_business_model.py
CSBS business case (titles sheet)	Final operational scenario: one 24 m² tilted bay is cleaned after controlled RGB detection; manual transfer between bays is explicitly charged as human labour. No site-specific PV generation, water saving, electrical-yield recovery or commercial payback is established. The three inherited numerical claims in Section 1 are deleted.
Constants hard-coded in the script today	Commercial inputs are explicitly labelled assumptions in final_business_inputs.json and varied in sensitivity.csv. Throughput/mechanical-work inputs come from the named actual simulation dataset. Modelled blower/auxiliary wattages and efficiency are assumptions; they are not electrical measurements.
Owner	CSBS Business & Techno-economic Systems Analyst: E058. Tags in this section carry that roll number.

Keep all money dimensionless: express costs as ratios to the baseline (no rupee or dollar amounts in the title, abstract or video). Every input needs a source or must be a clearly labelled assumption that you vary in the sensitivity graph.

3.1 Business model: formulas, your inputs and your results

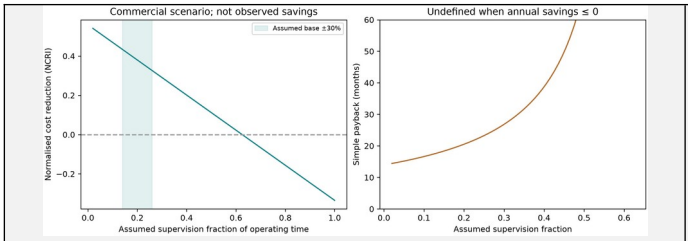
Quantity	Formula (define every symbol in your script)	Inputs used, with source or 'assumption'	Result (with unit)
Labour reallocation H_reclaimed	Hreclaimed = H0 − [f·n·Ta/(3600p) + n·(ttransfer + (1−p)·trecovery/p)/3600 + Hmaintenance]. n=sections/month; Ta=mean attempt seconds; p=success fraction; f=supervision fraction; all transfer/recovery is human labour in the base scenario.	All 30 proposed attempts at 20°, failures retained. H0=120 h/month; 21 sections/day × 30 days; f=0.2; attempt mean 612.713 s; p=1; transfer=180 s/section fully human; recovery=180 s/failure fully human; maintenance=10 h/month. All commercial values are assumptions.	57.055 person-h/month; robot-related human effort 62.945 h/month (monitoring + transfer/recovery + maintenance).
Payback horizon P_payback (dimensionless)	P_payback = CapEx / (T_life · ΔOpEx_annual); P < 1 ⇒ pays back within life T_life. Also report T_payback = CapEx / ΔOpEx in months.	Normalise one year of baseline labour cost to 1. CapEx ratio k=0.65; life L=5 years; annual saving ratio s=NCRI. All commercial inputs assumed. T=12k/s months; P=k/(L·s); undefined for s≤0.	T=20.522 months; P=0.34204 (dimensionless). Scenario output, not observed payback.
Normalised Cost Reduction Index NCRI	NCRI = 1 − κ; κ = annual robot-system cost / annual manual baseline cost. Costs are normalised to one baseline year. NCRI can be negative; payback is undefined when annual savings ≤ 0.	κ=human-hours/H0 + annual energy cost ratio + 0.05 spares/battery + 0.03 consumables. Electrical estimate: positive simulated work/assumed 0.60 efficiency + 20 W modelled blower + 5 W electronics; two assumed 3 W cameras. Tariff ratio=0.0003 per kWh of one-year baseline cost.	κ=0.61993; NCRI=1−κ=0.38007. Estimated system energy 51.281 kWh/year under scenario assumptions; not measured electrical energy.
Queuing delay	M/M/1: panel rows awaiting cleaning (λ per hour) served by one crawler (μ rows per hour). ρ = λ / (c·μ) < 1; M/M/1: W_q = ρ / (μ − λ); M/M/c: W_q = C(c, λ/μ) / (c·μ − λ) (Erlang C)	A section is one 24 m² bay, not a raster lane. λ=2.625 sections/h; μ=3600/[E(Tattempt)/p + (1−p)·180/p + 180]=4.54137 sections/h; c=1. Poisson arrivals/exponential service assumed.	ρ=0.57802; Wq=18.097 min if stable. Mean service 792.713 s. M/M/1 distribution assumptions are not validated by the simulator.
Sensitivity analysis	Vary the most uncertain input ±30 % (or across your scenario range) and plot P_payback and NCRI; report the break-even value.	Eight inputs swept ±30%: supervision, baseline hours, CapEx, manual transfer, failure recovery, energy price ratio, maintenance and drive efficiency. Repeated independent retries are an assumption. See sensitivity.csv; no PV yield uplift is monetised.	Break-even supervision fraction f=0.62536; maximum k for payback within assumed life=1.9004. Negative NCRI and unstable queues are retained, not converted to finite payback.

Queuing model used	M/M/1
λ, μ, c values used (with units)	λ=2.625 sections/h; μ=4.54137 sections/h; c=1; ρ=0.57802
Payback in months (T_payback)	20.5223 months; assumed scenario
Script output reproduces every number above when re-run	Yes

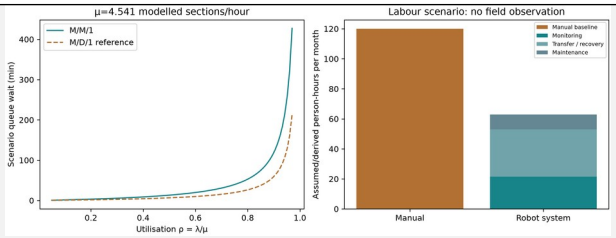
3.2 FMEA: failure modes, effects and risk priority (RPN = S × O × D, each 1–10)

Two failure modes are seeded from your project; add at least two more. Score before and after your mitigation (design change, control limit, sensor, procedure). The post-mitigation RPN must be justified by something you can show in the simulation or the code.

Failure mode	Effect on mission / business	S	O	D	RPN pre	Mitigation (what you changed)	RPN post	Shown in
Crawler slides down-slope on dusty glass (loss of traction)	Fall/damage, interrupted work and human recovery.	9	4	8	288	Physical containment guard, bounded return drive, attitude/height stop and logged guard load; guards remain part of the safety model. Post D=3 credits simulation fault detection only.	108	Both
Brush scratches anti-reflective coating	Permanent PV damage and repair cost.	8	4	9	288	Rounded soft-contact brush and load logs support development. Coating wear, grit hardness and allowable local pressure are unvalidated. No reduction credited: post S/O/D unchanged.	288	Not yet
RGB misses dirt or falsely declares clean	Residual dirt or wasted cleaning cycle.	6	5	8	240	Persistent two-frame image trigger; actual camera feeds; independent final residual check rejects false-clean development cases. Post D=3 is offline evaluation detectability, not field camera reliability.	90	Both
Large solid escapes or jams the guide	Unsafe ejection, jam or incomplete mission.	7	4	7	196	Permanent tray, fixed guide and actual solid/contact logs; success requires final retention. Applies only to the tested lower-strip corridor. Post D=3 credits final logged check.	84	Both
Dock transfer/accounting fault	Dirt inventory error or missed service.	5	4	7	140	Settled docking, 8 s modelled service, explicit mass balance and charge-energy counters. Post D=3 credits software checks; no airflow, connector or battery hardware reliability claim.	60	Both



Dimensionless commercial assumptions; no observed cost-saving claim.



Queue distribution assumed; transfer and maintenance count as human labour.

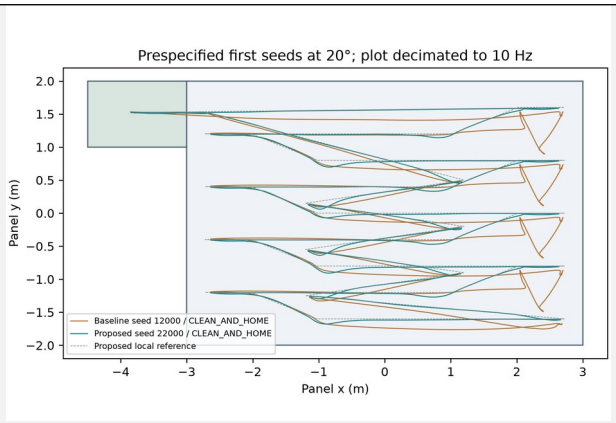
SECTION 4 · Simulation Telemetry, Monte Carlo Testing & Empirical Validation *Component 1 · CDIO Robotics (10)*

Evaluation protocol (titles sheet)	Revised evaluation: N=30/controller/tilt at 10°,20°,30° (180 full attempts), initial pose/soil-layout variation, complete-cycle success, footprint, motion/load, simulated mechanical work. Primary controller comparison: common first-30-second tracking RMSE. PV electrical restoration and measured electrical Wh/m² are not available.
Telemetry standard	Log at 500 Hz (MJCF timestep 0.002 s, or every 2nd step at 0.001 s): time, state (q, q̇ or pose, velocity), reference, tracking error, actuator torques/forces, FSM state, events. One CSV per run or one CSV with a run_id column.
Monte Carlo	N ≥ 30 runs per condition (more if your protocol says so). Randomise disturbances and initial conditions from stated distributions; record every random seed.
Required measures	Tracking RMSE = $\sqrt{(\Sigma\|p - p_ref\|^2 / n)}$ on the planned boustrophedon (zig-zag) sweep path vs MuJoCo crawler position; control effort E = $\int \tau^2 dt$ (sum of $\tau^2 \cdot \Delta t$ over all actuators); your primary task metric from the protocol.
Statistics	Compare the proposed controller with a baseline condition: Student's / Welch's t-test (two-sided, $\alpha = 0.05$), Cohen's d = $(\bar{x}_1 - \bar{x}_2) / s_pooled$, and a 95 % confidence interval.
Where real trial data goes	Group_06_Arush_Ronit_Harshvardhan/analytics/*_real_trials.csv, produced by your controller. Files named SIMULATED_* are starter targets, never results.
Owner	Telemetry QA Lead: E052.

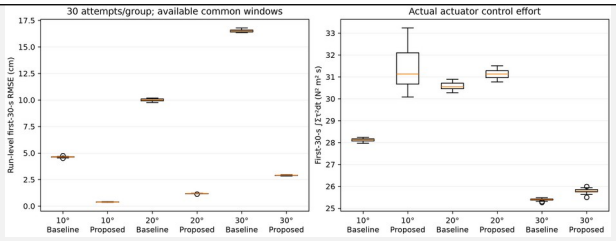
Baseline condition you compare against	Final collection model: waypoint reference + P wheel control versus segment lookahead + PI. Same camera pipeline, fixed brush/guide, mission route, guard, safety and shared incline/return regulator. Primary endpoint is common 0–30 s RMSE; all attempts continue through full mission/abort/1000 s.
Monte Carlo runs per condition (N)	30 independent runs/controller/tilt; 10°,20°,30°; 180 full-mission attempts. All failures retained. Table below: prespecified nominal 20°; all conditions in Appendix A.
Randomised parameters and their distributions	Initial x/y independent U(−0.01,+0.01) m; dust shape/offset RNG independent through logged debris_seed; clusters and 120 g lower-strip corridor fixed. Dry friction fixed. Seeds/distributions frozen before execution.
Logging rate actually achieved (Hz)	500 Hz; each 0.002 s integration sample. XZ compresses CSV; final batch serialises numbers to 9 significant digits.
Random seeds recorded for every run	Yes
Real-trials CSV path(s) in the repo	analytics/final_validation_v1/tilt_{10,20,30}/{baseline,proposed}_{seed}/ telemetry.csv.xz; summary.csv, protocol.json, statistics.json and trace_audit.json. A run, not a 500 Hz sample, is the unit of inference.

Measure	N	Baseline mean ± SD	Proposed mean ± SD	Unit	Better? (direction)
Tracking RMSE	Available 30 s windows: 30 baseline / 30 proposed; 30 attempts each	0.100061 ± 0.0012	0.0118225 ± 0.000166	m; common 0–30 s	Lower is better
Control effort $\int \tau^2\,dt$	Available 30 s windows: 30 baseline / 30 proposed; 30 attempts each	30.5728 ± 0.162	31.1333 ± 0.201	$N^2 \cdot m^2 \cdot s$; 0–30 s	Lower is better
Complete safe, camera-clean, solid-retained, serviced home cycle	30 per controller at 20°	30/30 (100.00%)	30/30 (100.00%)	completion fraction	Higher is better

Test used	Welch's t-test
t statistic, degrees of freedom, p-value (primary metric)	20°: t=-400.1377, df=30.11906, p=1.127902e-57; Holm-adjusted p=1.127902e-57 across 3 tilt comparisons.
Cohen's d (primary metric)	-103.3151; pooled SD; proposed minus baseline
95 % confidence interval of the difference	[-0.08868864868975783, -0.08778807408256452] m for proposed minus baseline. Run-level Welch interval; other conditions in Appendix A.



Prespecified first seeds at 20°; raw logs retained at 500 Hz.



All final conditions; 30 independent attempts per controller/tilt.

DATA PROVENANCE DECLARATION (mandatory). Reporting generated target numbers as experimental results is research misconduct and scores zero for this component.

Every number in Section 4 comes from MuJoCo runs of OUR controller, logged by our code	Yes
No value comes from SIMULATED_* files, generate_paper_figures.py or any np.random 'target' data	Yes
Our plots re-generate from the *_real_trials.csv files in the repo	Yes
Every input in Section 3 has a source or is labelled as an assumption	Yes

SECTION 5 · LinkedIn Video Showcase, AI Verification Audit & Peer Review *Component 3 · Showcase & Viva (5)*

LinkedIn post URL	Not supplied
Video length (mm:ss)	Simulation segment: 03:21.52, including visible cameras and the business graph. Face introduction, conclusion and final edited presentation have not been supplied.
Video shows the live MuJoCo simulation (not slides)	Yes
Video shows our business sensitivity graph	Yes
Webcam picture-in-picture: every member appears and speaks	Not verified

5.1 AI Interaction Audit Log: three cases where AI gave wrong robotics or economic data

Examples: a wrong MuJoCo attribute or API call, an invented statistic, a correct paper given the wrong journal, an economic formula with the wrong units. Show how you checked each one with a textbook, the official documentation or the original paper.

#	AI tool	What we asked & what AI said	How we caught it	What was actually true (with source)	Error type	Verdict
AI-01	ChatGPT	Conditional import mujoco.viewer inside mission was expected to work, but shadowed the module name and raised UnboundLocalError before model construction.	Ran / tested the code or simulation	Use from mujoco import viewer as mj_viewer. Actual mission/integration tests execute model construction. See docs/AI_AUDIT.md and controller import. Development audit candidate: members must review before claiming personal verification.	Wrong code / MuJoCo API	Corrected
AI-02	ChatGPT	RMS distance to the next waypoint was initially labelled path-tracking RMSE.	Ran / tested the code or simulation	Measure distance to the active segment using projection; tests/test_geometry_economics.py checks a 0.2 m perpendicular error. Time-indexed tracking is not claimed. Development audit candidate: members must review before claiming personal verification.	Wrong number or statistic	Corrected
AI-03	ChatGPT	Computing source hashes at run completion could label an earlier executed model with later edits.	Ran / tested the code or simulation	Capture hashes at launch and preserve immutable source snapshots. Final batch protocol records launch hashes; executed source is frozen. See docs/AI_AUDIT.md. Development audit candidate: members must review before claiming personal verification.	Wrong code / MuJoCo API	Corrected

5.2 Peer review: watch both assigned LinkedIn videos and score them (0–2 per criterion)

Group reviewed	Problem	Physics	Business	Data	Clarity	Best moment (mm:ss)	One question for the viva
Group 07: Autonomous Mobile Manipulator for Hospital Housekeeping & Patient-Room Clutter Clearing	2*	2*	2*	2*	2*	—	How does the mobile manipulator detect contact with a patient or unexpected obstacle and stop safely?
Group 12: Autonomous Pipeline Surveillance UAV							How is an apparent pipeline defect distinguished from a shadow, and how is detection confidence validated?

*Group 07: provisional scores supplied by the student as an overall assessment. The assigned video has not been verified and no best-moment timestamp is claimed. Group 12 remains blank pending clarification of the student's reference to Group 10.

SECTION 6 · Individual Learning Record & Research Consent *unmarked · one row per member · answer honestly*

These answers do not change your marks. They help improve the course and are used in anonymised research only if you consent in the sign-off below. Commit share: your commits ÷ the group's commits in your folder (git shortlog -sn -- <folder>).

Member	Hours I spent (total)	My declared role matched my actual work (1 = not at all, 5 = fully)	I learned a skill outside my role (robotics ↔ business)	Scripts I mainly wrote (file names)	My commit share (%)	Share of my writing first drafted with AI	Wrong AI facts / code I personally caught
Arush Ashish Patil (E048)	≈32 h*			Lead implementation: model XML; crawler_cleaning_controller.py; vision_mission.py; panel_vision.py; run_final_validation.py; final_business_model.py. AI-assisted.			
Ronit Rajput (E052)	≈32 h*			Supported model-file development and business-model inputs; Arush led script implementation.			
Harshvardhan Sahi (E058)	≈32 h*			Led business/economic inputs for final_business_model.py; Arush led script implementation.			

*Approximate retrospective estimate, not logged hours: two months treated as eight shared weekends, assuming four project hours per person per weekend. Additional individual hours are not estimated here.

Member	Before this course I had used a physics simulator	The most valuable thing I learned (one line)	One thing I would change in this course (one line)
Arush Ashish Patil (E048)			
Ronit Rajput (E052)			
Harshvardhan Sahi (E058)			

Track A: how this dossier becomes your 4-page IEEE paper

I. Introduction	Section 1: title, verified baseline friction, governing standards
II. Related work	Section 1.1: verified sources only (5+ with DOIs)
III. System model	Section 2: MJCF model, FSM, equations, control law
IV. Techno-economics & evaluation	Sections 3–4: CSBS model, FMEA, Monte Carlo results, statistics
V. Discussion & limitations	Sim-to-real gap, assumptions in the business model, AI audit lessons
Target venues (titles sheet)	A: IEEE INDICON · B: IEEE CASE

Individual Academic Sign-off *every member answers and signs*

Answer each statement for yourself: (1) this work is original and every source and AI use is declared; (2) our model loads and our controller and business scripts run from the repository; (3) I understand our whole project and can explain my own part in the viva voce.

Member (Roll)	(1) Original work	(2) Code runs	(3) Viva-ready	Signature	Optional research consent*
Arush Ashish Patil (E048)				SIGN PERSONALLY	
Ronit Rajput (E052)				SIGN PERSONALLY	
Harshvardhan Sahi (E058)				SIGN PERSONALLY	

**Optional: may Dr. Nanade use anonymised data from this dossier and repository in education research publications? Your answer has no effect on your marks, and you can withdraw until results are published.*

Appendix A · Evidence scope and final evaluation

Technical sections are populated from the completed frozen validation and reproducible business scenario. This copy uses ordinary editable text. Personal entries and peer-video reviews must be completed from the members' answers and the actual videos before submission.

Validated development movie (not independent statistical replication): CLEAN_AND_HOME at 610.858 simulated seconds; 32/32 selected irregular deposits below the residual threshold; qualifying whole-panel footprint 89.97917% (the exact-90% flag is false). Maximum rise above initial normal height 0.408684 mm; maximum tilt 0.093728°. Physical 120 g block leaves glass and is retained by the permanent tray. Initial dust inventory 4 g; final station 3.985638 g; residual glass 14.3625 mg; mass balance error 8.67e-19 kg. Final recording and headless nominal telemetry match exactly.

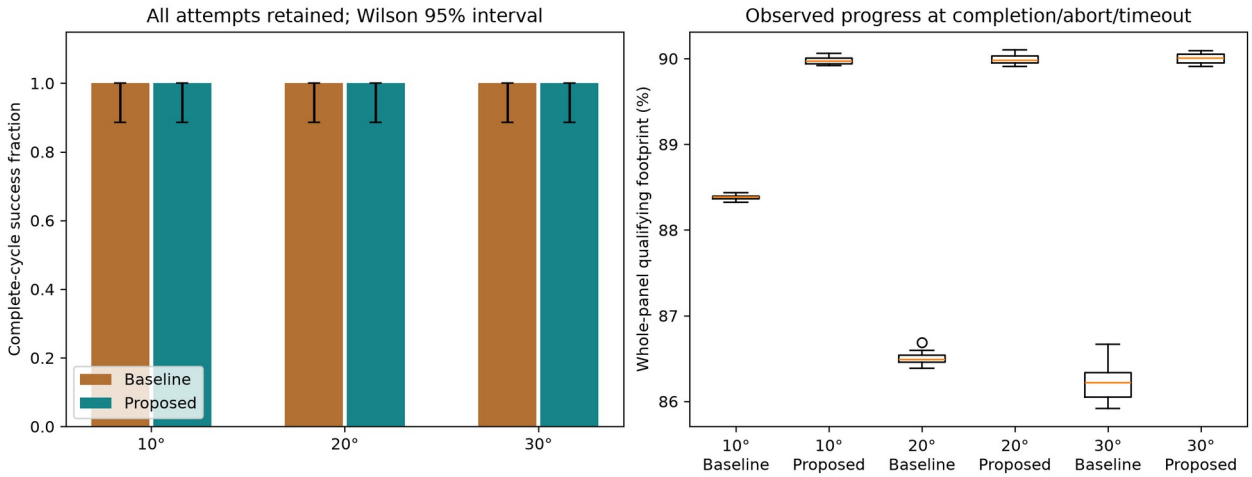
Tilt	Controller	Complete cycles	0–30 s RMSE mean ± SD (m)	Coverage mean (%)	Observed duration mean (s)
10.0°	baseline	30/30	0.0463282 ± 0.000453	88.3802	620.954
10.0°	proposed	30/30	0.00372785 ± 0.000159	89.976	610.025
20.0°	baseline	30/30	0.100061 ± 0.0012	86.4955	747.538
20.0°	proposed	30/30	0.0118225 ± 0.000166	89.9885	612.713
30.0°	baseline	30/30	0.165154 ± 0.00114	86.2146	937.735
30.0°	proposed	30/30	0.0290058 ± 0.000328	90.0024	867.773

10.0°: proposed−baseline Δ=-0.04260033 m; 95% CI [-0.04277804142081319, -0.04242262545673142]; Welch t=-486.1776, df=35.99813, p=2.579979e-70, Holm p=7.739937e-70, Cohen d=-125.5305. Completion Fisher p=1.

20.0°: proposed−baseline Δ=-0.08823836 m; 95% CI [-0.08868864868975783, -0.08778807408256452]; Welch t=-400.1377, df=30.11906, p=1.127902e-57, Holm p=1.127902e-57, Cohen d=-103.3151. Completion Fisher p=1.

30.0°: proposed−baseline Δ=-0.1361482 m; 95% CI [-0.13658916472867358, -0.1357072431479235]; Welch t=-627.6386, df=33.74324, p=3.260053e-70, Holm p=7.739937e-70, Cohen d=-162.0556. Completion Fisher p=1.

Full-run observed RMSE, effort and duration have differing exposure when a run aborts or times out; they are secondary descriptive metrics. All attempts, statuses, independent seeds, raw data and source snapshots are retained. Completion Wilson intervals and raw-trace audits are in statistics.json/trace_audit.json. No controller is declared superior solely because its mean is smaller; use the reported test and scope.



Engineering limitations and interpretation

Real RGB images drive contamination detection, persistence, targeted slowdowns/revisits and the departure decision. Navigation uses simulator ground-truth pose. Controlled colour/lighting segmentation is not a trained general debris detector. Two actual overhead sensor cameras and the onboard camera have visible equipment representations; scenery is isolated from validated sensor RGB. Clean/subthreshold development tests remain at home. General coloured/wet/shadowed dirt and arbitrary solid collection are unvalidated.

Dust removal is a contact-gated exponential layer/mass proxy, transferred to a modelled onboard bin and then station; there is no CFD vacuum or granular dust transport. The larger block has actual gravity/collision pushing, but its release corridor is prescribed. Four driven rollers approximate tracks. Zero-friction physical guards provide containment; they are not proof of real-world grip or safe operation on an

unguarded panel. Soft contact permits small penetration. Scratch, wear, motor, battery, structural mounting and electrical charging safety require hardware tests. The modelled 30 W dock service credits energy over eight seconds; it is not a demonstrated full recharge.

FMEA scores are ordinal engineering judgements, not fitted failure rates. Severity and occurrence stay unchanged; any post-RPN reduction credits only the specified code/logged evaluation detection. No reduction is credited to untested coating protection. The business case assumes one cleaner repeatedly services separately transferred 24 m² bays. 504 m²/day is a workload scenario, not an inferred 100 kWp plant. Energy efficiency/auxiliary loads are assumptions; PV electrical yield is neither simulated nor monetised.

Appendix B · References and source verification

The five entries below have bibliographic metadata verified in official or author-maintained software repositories. This does not imply access to all full articles or verification of a numerical PV-loss claim. The original authorised title is preserved as a question; its 15–18% phrase is not established by this work. See submission/source_audit/source_access.json for pinned commits, hashes and access scope.

[1] E. Todorov, T. Erez and Y. Tassa, “MuJoCo: A physics engine for model-based control,” IROS, pp. 5026–5033, 2012. DOI: 10.1109/IROS.2012.6386109. MuJoCo simulator context; developer README citation metadata verified.

[2] W. F. Holmgren, C. W. Hansen and M. A. Mikofski, “pvlib python: a python package for modeling solar energy systems,” JOSS 3(29), 884, 2018. DOI: 10.21105/joss.00884. PV modelling software context; official README metadata, not a PV-yield calculation in this project.

[3] K. S. Anderson et al., “pvlib python: 2023 project update,” JOSS 8(92), 5994, 2023. DOI: 10.21105/joss.05994. PV modelling context; official README metadata verified.

[4] M. G. Deceglie, L. Micheli and M. Muller, “Quantifying Soiling Loss Directly From PV Yield,” IEEE Journal of Photovoltaics 8(2), pp. 547–551, 2018. DOI: 10.1109/JPHOTOV.2017.2784682. Yield-based soiling measurement is separate from a dust-removal proxy; author-maintained rdtools README metadata only, full article not accessed.

[5] D. C. Jordan et al., “Robust PV Degradation Methodology and Application,” IEEE Journal of Photovoltaics 8(2), pp. 525–531, 2018. DOI: 10.1109/JPHOTOV.2017.2779779. Yield/degradation analysis context; author-maintained rdtools README metadata only, full article not accessed.

MuJoCo developer documentation (official source, pinned in source_access.json): contact friction coefficients, geom priority and solref/solimp interpretation. IEC 61215/61730 licensed clauses were not accessed; no invented clause or certified glass-load limit is used. Historical literature scaffolds and SIMULATED_* starter outputs are excluded from final evidence.

Appendix C · Reproduction and remaining team actions

From the group directory, use the pinned Python environment in requirements.txt. Headless execution requires MUJOCO_GL=egl before importing MuJoCo. Commands: submission/REPRODUCE.md. Raw final CSVs use XZ. This derivative uses ordinary editable text; original controls and choices remain in the preserved faculty Word form. Every original choice and answer is recorded in the companion FIELD_AUDIT.json.

Before submission, confirm the personal entries, complete the actual assigned video reviews, insert the final team video URL and obtain each member’s own signature. Faculty and Teams submission are separate.

Team contributions

Arush (E048) led the 3D modelling, coding, file implementation and MuJoCo simulation. Ronit (E052) supported 3D modelling and business modelling. Harshvardhan (E058) focused on business modelling and economics. The members worked together on weekends over approximately two months. Software development was AI-assisted.

Current demonstration: submission/complete_camera_video_v1/Group06_Complete_Simulation_With_Cameras.mp4 (3:21.52). The companion three-person narration script is ready, but actual face/voice footage and a published LinkedIn URL have not been supplied.